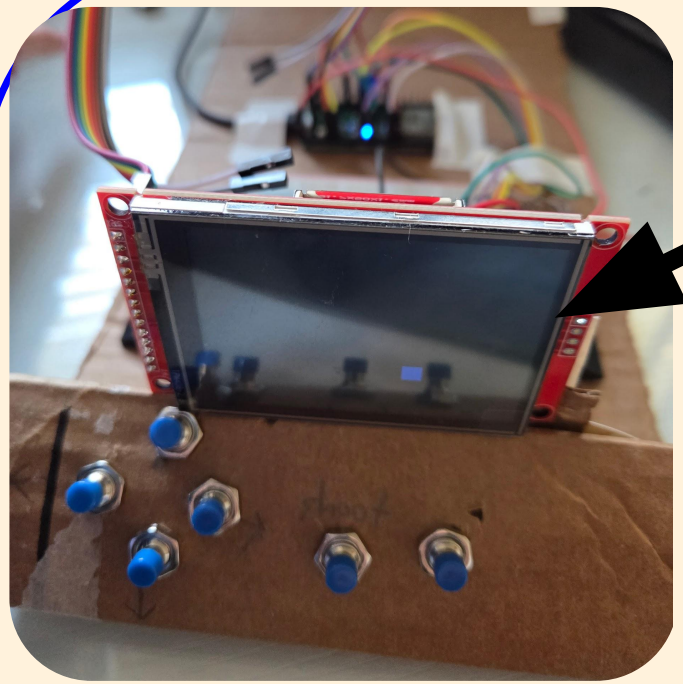
Arcade Game

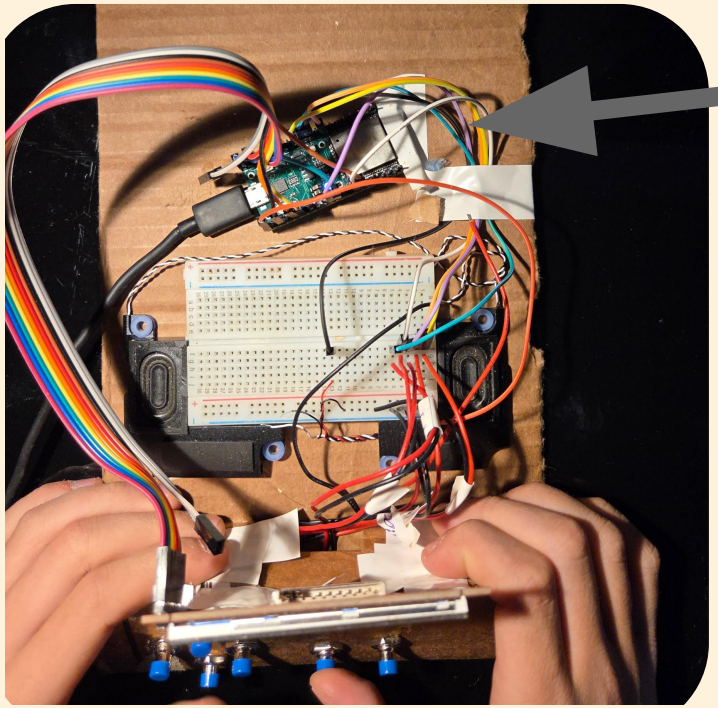
Project timeline

Assemble the electronics → Program the videogame → Cardboard box prototype → Wooden box



display

We connected a microcontroller to a display and buttons



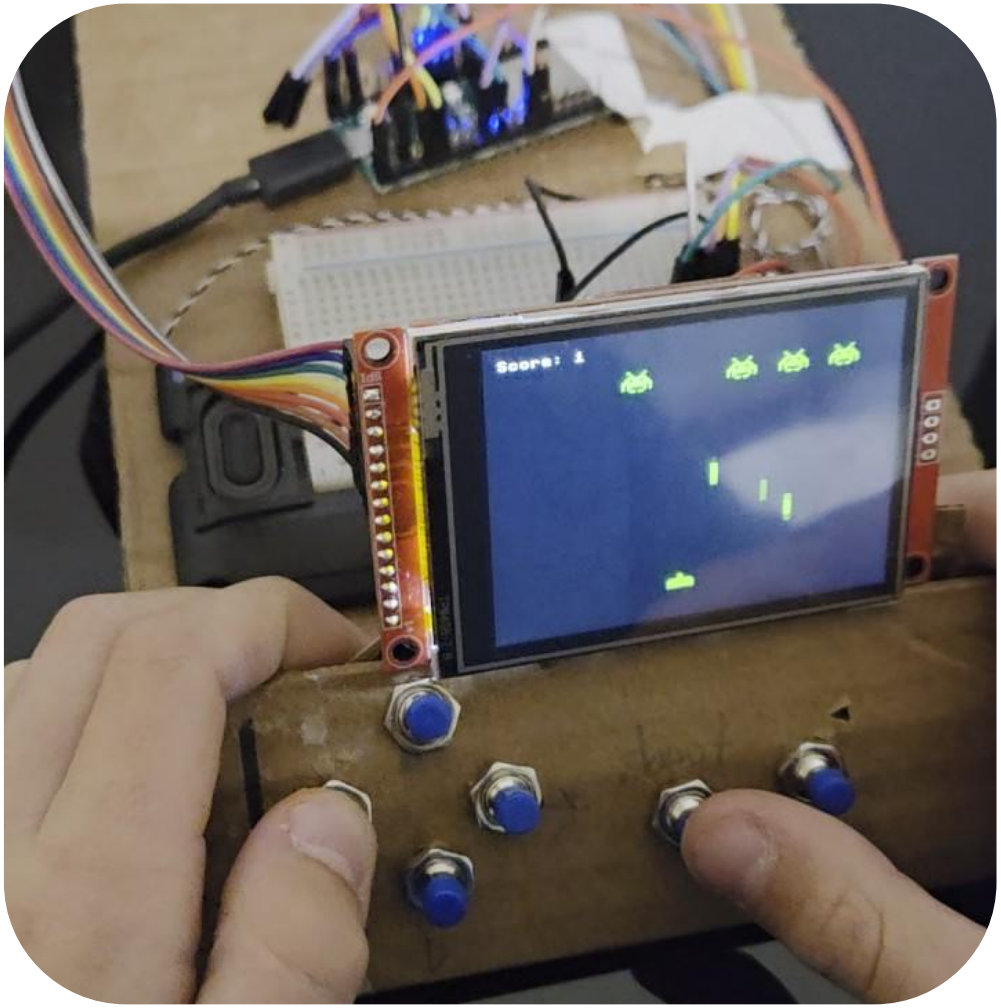
microcontroller

```python
while True:

    btn_fire.update()
    if game_state == 'playing':
        ### ============= PLAYER
        # update the player
        player.move()
        abc = 0

        ### ============= PLAYER's LASERS
        # fire (Create a Laser Sprite) while button pressed.

        if btn_fire.is_held():
            if time.time_ns() > last_laser_time + wait_before_new_laser:
                laserslist.append(Laser(display, x= round(player.x + player.w / 2 - 2.5), y=
                sfx.play_shoot()
                last_laser_time = time.time_ns()

        # update Lasers
        for l in laserslist:
            l.move()

        # check if any laser hit any invader. If an invader is hit, increase the score and d
        for l in laserslist:
            for i in invaders_list:
                if i.check_collision(l):
                    l.delete()
                    scoreboard.score +=1
                    scoreboard.draw()
                    i.delete()

        # remove inactive lasers
        laserslist = [l for l in laserslist if l.active]

        if scoreboard.score >= score_to_win:
            game_state = 'you_won'

        ### ============== INVADERS
        #update invaders
        Invader.move_row(invader_speed)
        for invader in invaders_list:
            invader.move(invader_speed)
```

I coded the videogame with micropython.

I coded the player, invaders and lasers and a loop to make them move

https://github.com/simonemerlin/esp32arcade

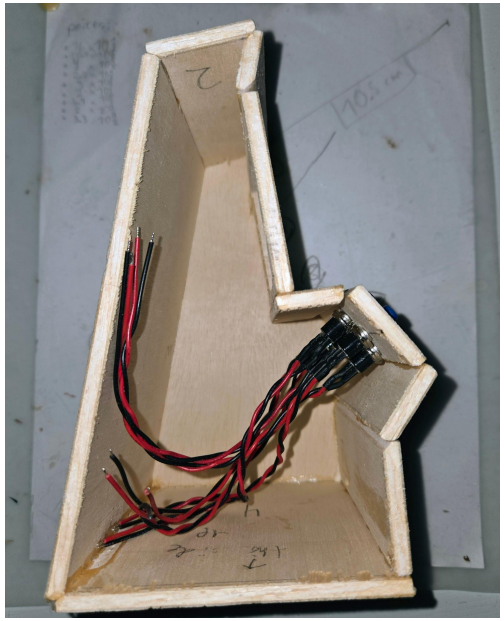


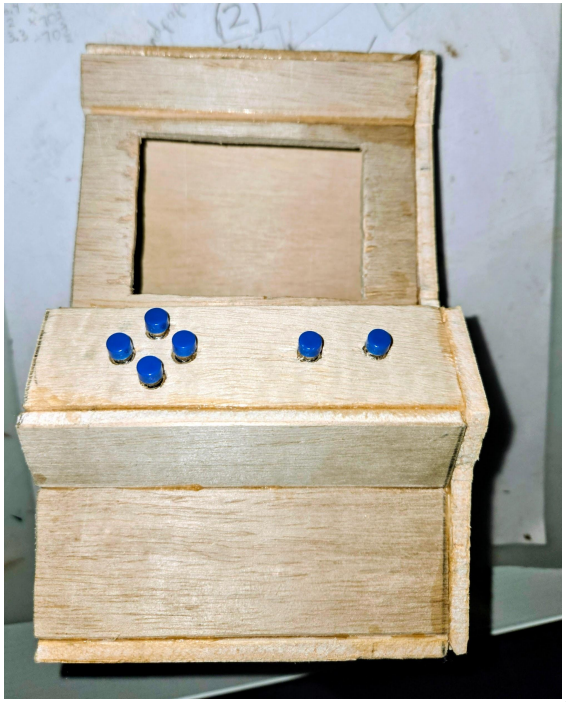
This is the first prototype working

I built a cardboard prototype for the box.

This is the screen hole for the screen

The buttons for the game will go there





This is the final product after we cut out all the wood pieces and glued them together.

What I learned

How to code in micropython

How to build prototypes

How to fix bugs in coding

How to build with wood